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30 March 1961
SUPERSEDING
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31 July 1957

MILITARY SPECIFICATION

TRANSMITTER, TARGET DESIGNATION, MARK 24 MODS 0 and 1, and CONTROL UNIT MARK 79 MODS 0 and 1

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE.

1.1 Scope. - This specification covers the requirements for procurement of Target Designation Transmitter Mark 24 Mod 0 and Control Unit Mk 79 Mod 0; and, Target Designation Transmitter Mark 24 Mod 1 and Control Unit Mk 79 Mod 1.

2. APPLICABLE DOCUMENTS.

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

FEDERAL

UU-P-271 - Paper, Wrapping, Waterproofed Kraft.
PPP-B-601 - Boxes, Wood, Cleated Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-A-140 - Adhesive, Water Resistant, Waterproof Barrier Material.
MIL-T-152 - Treatment, Moisture and Fungus Resistant, of Communications, Electronic, and Associated Electrical Equipment.
MIL-A-5092 - Adhesive, Rubber (synthetic and reclaimed rubber base).
MIL-I-11748 - Interference Reduction for Electrical and Electronic Equipment.
MIL-G-12803 - Gasket Materials, Non-metallic.
MIL-G-15793 - Grease, Instrument.
MIL-I-16910 - Interference Measurement, Radio, Methods and Limits; 14 Kilocycles to 1000 Megacycles.

FSC 1265

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MIL-T-17113 - Tests; Shock, Vibration, and Inclination (for Electronic Equipment), General Specification.
MIL-F-18870 - Fire Control Equipment, Naval Ship and Shore: General Specification.
MIL-W-21927 - Weapons, Handling and Preparation for Delivery of; General Requirements for.
MIL-I-26860 - Indicator, Humidity, Plug, Color change.

STANDARDS

MILITARY

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures of Electric and Electronic Equipment.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.

DRAWINGS

BUREAU OF NAVAL WEAPONS

LD 412563 - Target Designation Transmitter Mark 24 Mod 0 (Assembly) List of Drawings, Assemblies, Parts, Specifications, Etc.
LD 416780 - Target Designation Transmitter Mark 24 Mod 1 (Assembly) List of Drawings, Assemblies, Parts, Specifications, Etc.

PUBLICATIONS

BUREAU OF NAVAL WEAPONS

NAVORD OD 16420 - Quality Assurance Provisions for Target Designation Transmitter Mark 24 Mod 0.
- Quality Assurance Provisions for Target Designation Transmitter Mark 24 Mod 1.

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

INTERSTATE COMMERCE COMMISSION
Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS.

3.1 Preproduction sample. - Unless otherwise specified, a preproduction sample is required for each contract or order. Samples shall be manufactured using the methods, procedures, and plant location proposed for the production lot. Samples shall be inspected as specified in 4.4.4.1; such inspection being for the purpose of determining that the manufacturer's production process will produce target designation transmitters and control units that will meet the requirements of this specification.

3.2 Functional. - The target designation transmitter and control unit shall perform the following functions:

(a) Transmit and display optical bearing, elevation, and estimated target range information.
(b) Provide a switch control for selecting fire control directors to which the optical tracking information may be transmitted; and indicate by means of lights, the directors selected.
(c) Indicate director status by means of lights.

3.3 Construction. - The target designation transmitter and control unit shall be manufactured in accordance with LD 412563 or LD 416780, as applicable; and, shall meet all requirements specified thereon, and on all referenced drawings, documents and publications.

3.3.1 Cleanliness. - The interiors of the target designation transmitter and control unit and all components therein shall be clean and free from oil, dust, dirt, and foreign matter of any kind.

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3.3.2 Mechanical operation. - The training and elevating assemblies, all hand operated cranks and knobs, and all rotating parts shall operate freely without binding or encountering high spots.

3.3.3 Wiring. - Electrical wiring shall be in accordance with the wiring diagrams listed on LD 412563 or LD 416780, as applicable. Where necessary, wires shall be properly secured so as to prevent interference with moving parts.

3.4 Power. - The target designation transmitter and control unit shall operate satisfactorily when applicable units of the equipment are supplied with the following power:

115V ± 10%, 60 cycle ± 15%, 1 phase (Mod 0 only)
115V ± 10%, 400 cycle ± 5%, 1 phase
60V ± 5%, DC
-60V ± 5%, DC
+48V ± 10%, DC

3.5 Transmission. -

3.5.1 Train and elevation. - The transmission error in train and elevation shall not exceed 1 degree of arc at 1 speed as measured between the binocular cross-line reticle and the synchro receiver dials in the control unit and test panel.

3.5.2 Range. - The transmission errors in range shall not exceed the following limits:

Control Unit Mk 79 Mod 0

Yds per rev	Max Allowable error
2,000	50 yds
72,000	500 yds
1,296,000	5,000 yds

Control Unit Mk 79 Mod 1

Yds per rev	Max allowable error
2,000	50 yds
72,000	500 yds
144,000	1000 yds
1,296,000	5,000 yds

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3.6 Servo-loop. - (See 6.4 for definitions).

3.6.1 CT gain limit. - The requirements of 3.6.2 through 3.6.4 shall be met with a maximum CT setting of 90.

3.6.2 Constant velocity error coefficient. - The constant velocity error coefficient shall be 80 per second minimum with an input velocity of 75 degrees per second.

3.6.3 Overshoot limit. - The overshoot, as measured by the error voltage, shall be less than 33 percent when the servo-loop is given a 1.0 volt step function input.

3.6.4 Static error limit. - The static error, measured across terminals R1 and R2 of the control transformer, shall not exceed 0.05 volt rms.

3.6.5 Sine-cosine POT requirements. - The sine-cosine POT shall meet the following requirements with the servo loop energized and the range input to the POT set at 32 volts:

Function	Requirement
Sine Output	$32 \sin \theta \pm 1$ volt
Cosine Output	$32 \cos \theta \pm 1$ volt

Where θ = traverse angle of the target designation transmitter.

3.7 Dielectric strength. - The electrical components of the target designation transmitter and control unit shall withstand the dielectric test voltages specified in Specification MIL-F-18870.

3.8 Insulation resistance. - The insulation resistance of each electrical circuit of the equipment shall be as specified in Specification MIL-F-18870.

3.9 Environmental requirements. -

3.9.1 Shock. - The assembled target designation transmitter and control unit shall be capable of withstanding Type I, Class B shock as specified in Specification MIL-T-17113.

3.9.1.1 Shock-transportation, storage and handling. - The assembled target designation transmitter and control unit, after being subjected to the tests specified in 4.4.4.3.1.1, shall show no damage and shall meet all the requirements specified herein.

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3.9.2 Vibration. - The assembled target designation transmitter and control unit shall be capable of withstanding Type I environmental vibration as defined in Standard MIL-STD-167, at frequencies between 200 and 1500 cycles per minute at an amplitude of plus or minus 0.030 inch.

3.9.2.1 Vibration-transportation. - The assembled target designation transmitter and control unit, after being subjected to the tests specified in 4.4.4.3.2.1, shall show no damage and shall meet all the requirements specified herein.

3.9.3 Temperature extremes. - The target designation transmitter and control unit shall meet the requirements of this specification when the equipment is at minus 20°F and also at plus 160°F. Stabilization at these temperatures or the changes in temperature involved shall not change any performance characteristic of the equipment.

3.9.4 Watertightness. - The assembled target designation transmitter and control unit shall be watertight when tested in accordance with 4.4.4.3.4.

3.9.5 Radio interference suppression. - Radio interference limits and reduction of radio-frequency interference emanation and susceptibility to such emanation shall be in accordance with Specifications MIL-I-11748 and MIL-I-16910, as applicable.

3.9.6 Moisture and fungus resistance. - The requirements of Specification MIL-T-152 shall apply to nonmetallic parts and materials subject to moisture and fungus attack.

3.9.7 Corrosion protection. - During the course of manufacture the surfaces of all parts and assembled units shall be protected against corrosion prior to and after assembly.

3.10 Detailed requirements. -

3.10.1 Target designation transmitter. -

3.10.1.1 Tracking head height. - Rotation of the height adjusting handwheel shall raise or lower the tracking head a total of 10 inches between maximum and minimum height. Overall travel shall be smooth without binding. The protective boot around the pedestal stanchion shall fold and unfold freely and uniformly without damage.

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3.10.1.1.1 Height adjusting locking device. - Rotation of the height locking handwheel shall operate to lock or release the pedestal support shaft. The handwheel shall rotate freely without binding or sticking.

3.10.1.2 Tracking head rotation. - The tracking head assembly shall be free to rotate through 320 ± 5 degrees in train. The head shall rotate at a constant rate without tight or loose spots. Torque required to train the head assembly shall not exceed 8 foot-pounds.

3.10.1.2.1 Train locking device. - Insertion of a train locking pin into any one of the seven holes in the locking ring shall lock the tracking head assembly in train. Insertion and withdrawal of the locking pin shall be free from binding or sticking.

3.10.1.2.2 Train scale. - When the train locking pin is engaged in the forward position, as indicated by the vertical arrow on the pedestal, the scribed index affixed to the tracking head shall indicate 0 ± 30 minutes on the stationary circular scale located above the handrail.

3.10.1.3 Elevation. - The elevating (left) arm assembly shall be free to rotate through 90 degrees in elevation and 20 degrees depression, within a tolerance of ± 2 degrees. The elevating arm assembly, with all equipment installed, shall balance at 0 degrees elevation when a load of 8 ± 1 pound is applied to the switch grip.

3.10.1.3.1 Elevation locking device. - Insertion of an elevation locking pin into the hole in the damper cover plate shall lock the elevating arm assembly at 0 degrees elevation. Insertion and withdrawal of the locking pin shall be free from binding or sticking.

3.10.1.3.2 Elevation scale. - When the elevation locking pin is engaged, the elevation indicator mounted on the elevating arm shall indicate 0 ± 30 minutes on the dial secured to the damper housing.

3.10.1.4 Left (elevating) arm assembly. -

3.10.1.4.1 Binocular mounting assembly. - The detent that secures the binocular holder shall engage without binding when the assembly is pushed forward. Each binocular mounting cylinder shall be operable through the full interpupillary travel without interference. The head rest assembly shall pivot freely to and from the operating position and the spring shall lock the detent in both positions.

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3.10.1.4.1.1 Horizontal parallelism. - The maximum allowable deviation from the binocular line of sight, referenced to a plane parallel to the mounting flange of the pedestal, shall be ± 5 minutes of arc throughout the entire range of train of the tracking head assembly.

3.10.1.4.1.2 Vertical parallelism. - The maximum allowable deviation of the cross-line reticle in the binoculars, referenced to a plane perpendicular to the mounting flange of the pedestal, shall be ± 10 minutes of arc throughout the entire range of elevation and depression of the oscillating assembly.

3.10.1.4.2 Open sight. - The maximum allowable deviation of the line of sight of the open sight, referenced to a plane parallel to the binocular line of sight, shall be ± 15 minutes of arc.

3.10.1.5 Right arm assembly. - The right arm assembly shall be free to rotate through 90 degrees without interference.

3.10.1.5.1 Right arm assembly locking device. - A spindle locking device shall lock the right arm in any one of three positions. Rotation of the spindle handle shall be free from binding and sticking.

3.10.1.5.2 Press-to-talk switch grip assembly. - The switch grip assembly shall be free to pivot through 100 degrees without interference.

3.10.1.6 Lubrication. - The target designation transmitter shall be lubricated in accordance with BuOrd Drawing 1691611, covering a detail of LD 412563 and LD 416780.

3.10.2 Control unit. -

3.10.2.1 Estimated range crank and dial. - The range dial shall remain stationary in any position in which it is placed by the range crank under all environmental conditions.

3.10.2.2 Dials. - All dials shall rotate freely throughout their full range of rotation without binding or touching adjacent parts.

3.10.2.3 Lubrication. - The control unit shall be lubricated with instrument grease in accordance with Specification MIL-G-15793.

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3.11 Workmanship. - Workmanship shall be commensurate with good manufacturing standards for similar equipment. Workmanship shall be of such quality as to produce material of uniform high quality and as to meet all requirements of the applicable drawings and of this specification.

4. QUALITY ASSURANCE PROVISIONS.

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. -

4.2.1 Target designation transmitters and control units. - A lot shall consist of all the target designation transmitters and control units manufactured by the same process and at the same plant location and submitted for Government acceptance at any one time.

4.3 Sampling. -

4.3.1 Sampling for preproduction inspection. - Except as specified in 4.3.1.1, one preproduction sample target designation transmitter and one control unit, packaged and packed in accordance with Section 5, shall be supplied by the supplier as specified in 3.1. The sample shall be submitted to the Government inspector for preproduction inspection (see 4.4.4.1) at an activity designated in the contract or order (see 6.2) or by the Bureau of Naval Weapons. Further production of target designation transmitters and control units by the supplier prior to the approval of the procuring activity or completion of inspection of the preproduction sample shall be at the supplier's risk.

4.3.1.1 Waiver of preproduction sample. - If a supplier has previously delivered target designation transmitters and control units in accordance with the requirements of this specification

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and his product has been found to be satisfactory, the pre-production sample and inspection for any subsequent contract may be waived at the discretion of the procuring activity (see 6.2).

4.3.2 Sampling for periodic tests. - Unless otherwise specified in the contract or order (see 6.2), sample target designation transmitters and control units, packaged and packed in accordance with Section 5, shall be selected at random by the Government inspector, as indicated below, for periodic testing (see 4.4.4.2). The supplier shall forward the sample selected by the Government inspector to a testing activity designated in the contract or order (see 6.2) or by the Bureau of Naval Weapons. The sampling procedure shall be as follows:

(a) One transmitter and control unit from the first 25 or less.
(b) One transmitter and control unit from the next 50 or less.
(c) One transmitter and control unit from the next 75 or less.
(d) One transmitter and control unit from each succeeding quantity of 100 or less.

If a sample from any of the above quantities fails the periodic tests, sampling shall be repeated starting with (a) above.

4.3.3 Sampling for lot acceptance. - No provisions are made for sampling for lot acceptance since each target designation transmitter and control unit in each lot shall be subjected to the lot acceptance tests of 4.4.1.

4.4 Inspection. -

4.4.1 Lot acceptance tests. - Lot acceptance tests in accordance with Publication NAVORD OD 16420 or the Quality Assurance Provisions for Target Designation Transmitter Mk 24 Mod 1, as applicable, shall be performed on each target designation transmitter and control unit to determine conformance with the requirements of 3.1 through 3.8 and 3.10. Target designation transmitters and control units that fail to pass any lot acceptance test shall be rejected.

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4.4.2 Visual and dimensional examination. - Each target designation transmitter and control unit as specified in 4.3.3 shall be visually examined for completeness of manufacture, freedom from defects, proper identification, and workmanship. Dimensional and other nondestructive inspections shall be made in accordance with the acceptance inspection instructions provided by the procuring activity.

4.4.3 Preservation, packaging, packing, and marking. -

4.4.3.1 Target designation transmitters and control units. - Each target designation transmitter and control unit shall be examined to determine that preservation and packaging are in accordance with the requirements of Section 5. Each container and container marking shall be examined to determine conformance with the requirements of Section 5.

4.4.4 Tests and test procedures. -

4.4.4.1 Preproduction tests. - Preproduction target designation transmitters and control units (see 4.3.1) shall be subjected to the examinations of 4.4.2, the environmental tests of 4.4.4.3.1 through 4.4.4.3.5 and the lot acceptance tests of 4.4.1. Target designation transmitters and control units that pass the required tests and meet the requirements specified herein shall be included in the quantity called for in the contract or order.

4.4.4.2 Periodic tests. - Periodic tests (see 4.3.2) shall consist of the environmental tests of 4.4.4.3.1 through 4.4.4.3.5 and shall be performed on target designation transmitters and control units that have previously passed the examinations of 4.4.2 and the lot acceptance tests of 4.4.1.

4.4.4.2.1 Failures during periodic tests. - In the event of failure of any target designation transmitter or control unit during periodic testing, inspection and acceptance of subsequent target designation transmitters or control units shall be suspended until the reasons for failure and consequent rejection have been determined and the cause of failure corrected by the supplier to the satisfaction of the procuring activity. The supplier shall furnish the procuring activity with a written statement properly endorsed by the Government inspector indicating the reasons for failure and the corrective action taken.

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4.4.4.3 Environmental tests. -

4.4.4.3.1 Shock. - The assembled target designation transmitter and control unit shall be shock tested in accordance with the procedures of Specification MIL-T-17113 to determine conformance with 3.9.1.

4.4.4.3.1.1 Shock-transportation, storage and handling. - The assembled target designation transmitter and control unit, packaged and packed as specified in Section 5, shall be subjected to the appropriate rough handling tests of Specification MIL-W-21927 to determine conformance with 3.9.1.1.

4.4.4.3.2 Vibration. - The target designation transmitter and control unit shall be vibration tested to determine conformance with 3.9.2. The vibrations shall be applied separately in each of the three mutually perpendicular directions. Each test in each direction shall be of at least 15 minutes duration.

4.4.4.3.2.1 Vibration-transportation. - The assembled target designation transmitter and control unit, packaged and packed as specified in Section 5, shall be vibration tested in accordance with Specification MIL-W-21927 to determine conformance with 3.9.2.1.

4.4.4.3.3 Temperature extremes. - The target designation transmitter and control unit shall be tested at extremes of temperature to determine conformance to 3.9.3. The equipment shall be placed in a cold chamber and tested while at minus 20°F. The equipment shall also be placed in a hot chamber and tested while at plus 160°F. The sequence of hot and cold testing shall be at the discretion of the testing activity. When the equipment is stabilized at the cold and at the hot temperature extremes specified herein, tests shall be performed to the extent that the Government inspector is able to determine conformance with 3.9.3.

4.4.4.3.4 Watertightness. - The target designation transmitter and control unit shall be tested for watertightness to determine conformance to 3.9.4. The watertightness of the equipment shall be determined by conducting the floodtight test in accordance with Standard MIL-STD-108.

4.4.4.3.5 Radio interference suppression. - The target designation transmitter and control unit shall be tested for environmental radio interference and suppression to determine con-

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formance with 3.9.5. The tests shall be conducted in accordance with the applicable procedures of Specifications MIL-I-11748 and MIL-I-16910.

4.4.4.3.6 Moisture and fungus resistance. - The target designation transmitter and control unit shall be tested for moisture and fungus resistance to determine conformance with 3.9.6.

4.4.4.3.7 Corrosion protection. - Visual examination shall be made to determine conformance with 3.9.7.

4.4.4.3.8 Examination and retests following environmental tests. - Following the environmental tests of 4.4.4.3.1 through 4.4.4.3.5, the target designation transmitter and control unit shall be examined for breakage, loose parts, and other defects. Target designation transmitters and control units that pass the environmental tests shall be returned to the contractor for any needed repairs and reconditioning. The lot acceptance tests of 4.4.1 shall be repeated as necessary to determine that no changes in mechanical, electrical, and optical performance have taken place as a result of the environmental tests.

4.5 Resubmission. - Resubmission of rejected target designation transmitters and control units for retest shall be as specified in the contract or order (see 6.2).

5. PREPARATION FOR DELIVERY.

5.1 Required levels. - Target designation transmitters and control units shall be preserved, packaged, and packed in accordance with one of the following levels as specified in the contract or order (see 6.2).

5.2 Preservation. -

5.2.1 Level A. -

5.2.1.1 Cleaning. - All exterior surfaces of target designation transmitters and control units shall be cleaned in accordance with process C-1 of Specification MIL-P-116.

5.2.1.2 Drying. - After cleaning, any drying required shall be in accordance with procedure D-1 or D-4 of Specification MIL-P-116.

5.2.2 Level B. - Not applicable.

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5.2.3 Level C. - Cleaning, drying, and preservation of target designation transmitters and control units shall be in accordance with the manufacturer's commercial practice.

5.3 Packaging. -

5.3.1 Level A. - Target designation transmitters and control units shall be individually packaged in accordance with submethod IIa of Method II of Specification MIL-P-116. Movable parts susceptible to damage shall be secured, blocked, and braced to prevent movement prior to packaging. Material for gaskets used to fabricate the package shall conform to Specification MIL-G-12803, Type 1, Class 2, Group 1, Grade 1. The adhesive used for cementing the gaskets shall conform to Specification MIL-A-5092, Type III. The humidity indicator for the package shall conform to Specification MIL-I-26860, Type II.

5.3.2 Level B. - Not applicable.

5.3.3 Level C. - Target designation transmitters and control units shall be packaged for domestic shipment in accordance with the manufacturer's commercial practice. The packaging shall be adequate to provide protection from deterioration and physical damage during direct shipment from source of supply to the first receiving activity for immediate use.

5.4 Packing. -

5.4.1 Level A. - Target designation transmitters and control units shall be individually packed one each in an overseas type box conforming to Specification PPP-B-601 or PPP-B-621. Shipping containers shall be lined with a suitable class case liner material in accordance with Specification UU-P-271. Case liners shall be completely sealed with water resistant adhesive conforming to Specification MIL-A-140. Each container shall be provided with suitable rainproof vents. The interior of each container shall be provided with suitable blocking and cushioning so as to prevent shifting of and damage to items contained therein. Each container shall be modified to include a skid base with suitable skid. Each target designation transmitter container shall be provided with a suitable extended base, properly braced, so that the container with its contents will be stable in the intended upright position. The box shall be provided with an inspection port for viewing the humidity indicator.

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5.4.2 Level B. - Level B packing shall be identical with Level A packing except that packing containers shall be domestic type boxes conforming to Specification PPP-B-601 or PPP-B-621.

5.4.3 Level C. - Target designation transmitters and control units shall be packed or otherwise prepared for shipment to insure carrier acceptance and to insure safe delivery to destination at the lowest applicable rate. Containers shall comply with the Consolidated Freight Classification Rules or other carrier regulations applicable to the mode of transportation.

5.5 Marking of shipping containers. -

5.5.1 For production target designation transmitters and control units. - In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of Military Standard MIL-STD-129, including the serial number of the units contained therein.

5.5.2 For preproduction target designation transmitters and control units. - Shipping containers shall be marked as specified in 5.5.1 and with the following information:

Title, number, and date of this specification
Mark and Mod and LD number
Sample for preproduction inspection
Name of manufacturer
Manufacturer's plant address
Reference (contract number)

5.5.3 For target designation transmitters and control units for periodic testing. - Shipping containers shall be marked as specified in 5.5.2 except that containers shall be marked "Sample for periodic tests" instead of "Sample for preproduction inspection". In addition, containers shall be marked with the identification of the lot or quantity from which the sample was taken and the serial number of the sample.

6. NOTES.

6.1 Intended use. - The target designation transmitter and control unit covered by this specification are together intended for installation in the air defense station of naval vessels as an anti-aircraft target designation station.

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6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Testing activity for preproduction samples or samples for periodic tests if the activity is not designated by the Bureau of Naval Weapons (see 4.3.1 or 4.3.2).
(c) Whether preproduction samples and inspection shall be waived (see 4.3.1.1).
(d) Resubmission procedure for rejected target designation transmitters and control units (see 4.5).
(e) Levels required in preparation for delivery (see 5.1).
(f) Special marking when required (see 5.5.1).

6.3 Preparation for delivery criteria. - Criteria for use of the proper level of preservation, packaging, and packing are indicated below in 6.3.1 through 6.3.3.

6.3.1 Level A. - This level shall be used to afford adequate protection of items during multiple shipment and handling, indeterminate storage, and worldwide redistribution.

6.3.2 Level B. - This level shall be used for protection of items during multiple domestic shipment and handling and covered storage. Six month cyclic inspection and any necessary re-preservation and repackaging is required when this level is used.

6.3.3 Level C. - This level shall be used only when it is definitely known that items are to be shipped to domestic installations for immediate use at the first receiving activity.

6.4 Definitions. -

6.4.1 Control transformer gain. - Control transformer (CT) gain is the ratio of the output voltage measured across the control winding of the servo motor to the input voltage measured across the CT input terminals of the synchro control unit with the stabilizing potentiometer (STAB POT) in its full counter-clockwise position.

6.4.2 Constant velocity error coefficient. - The constant velocity error coefficient is the ratio of the constant velocity input in degrees per second to the displacement error in degrees. Since the control transformer has a gradient of one volt per degree, in the range from zero to ten degrees, the displacement

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error in degrees is numerically equal to the error voltage in volts rms measured across terminals R1 and R2 of the control transformer. Constant velocity input means that the input angle increases linearly with time.

6.4.3 Overshoot. - Overshoot is the amount the output shaft overshoots when the servo-loop is given a step function input. The overshoot is expressed as the percentage of the magnitude of the step function.

6.4.4 Static error. - Static error is the in-phase component of the error voltage that is necessary to overcome the static friction of the gear train and the bearings.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-6279

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DEPARTMENT OF THE NAVY

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NAVY DEPARTMENT

OFFICIAL BUSINESS

Commander, Naval Ordnance Systems Command
Maintenance Management and Supporting
Standardization Section
Attn: ORD-05223
Department of the Navy
Washington, D. C. 20360

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